

Spatial Data Operations

HES 505 Fall 2025: Session 10

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Goals for Today

- **Describe** the Modifiable Areal Unit Problem (MAUP).
- **Link** the inferential problems of the ecological fallacy to the statistical problems of the MAUP.
- **Modify** the geometries of spatial objects to explore the MAUP.

Problems of Inference

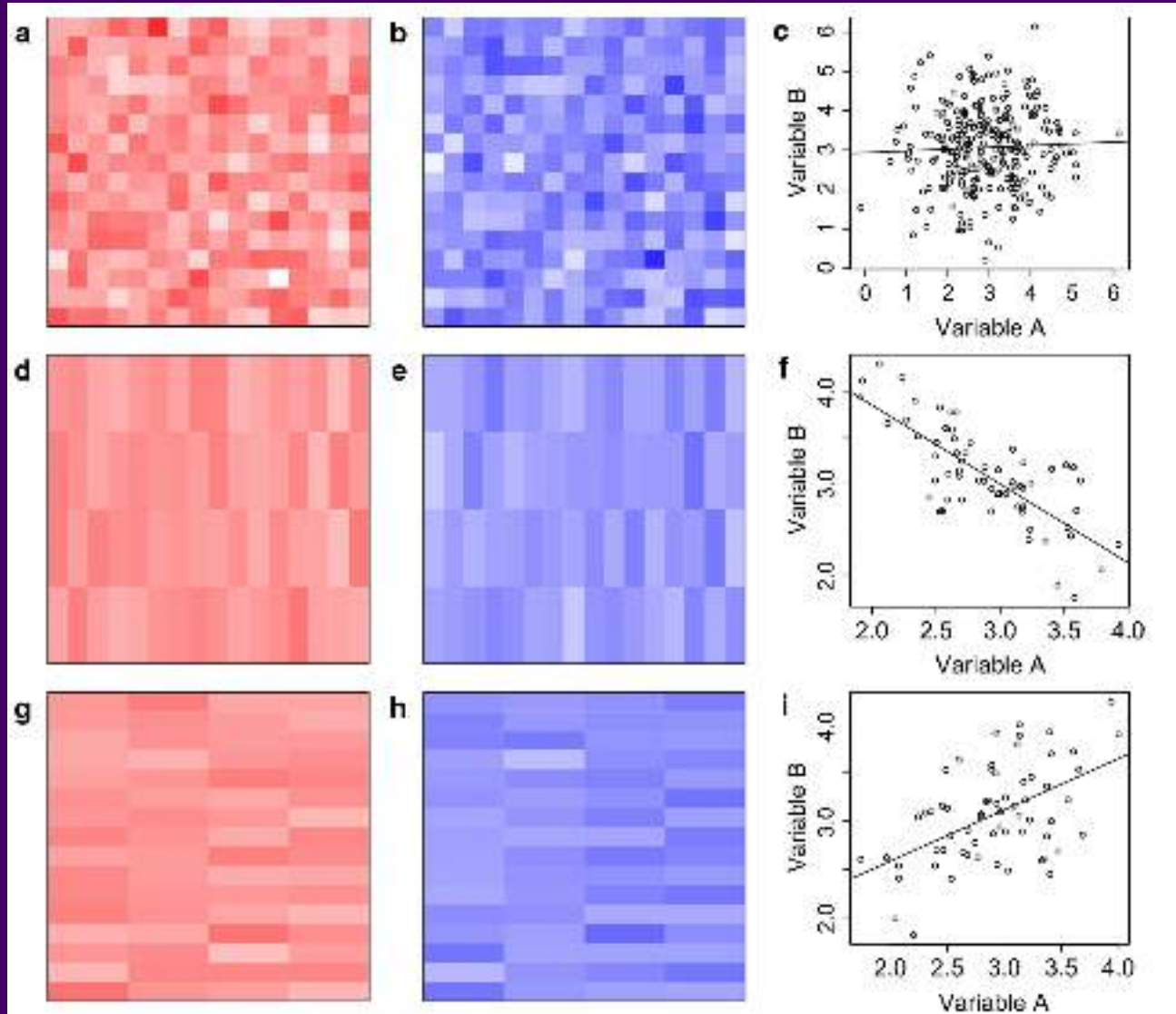
- Atomic Fallacy
- Ecological Fallacy

Problems of Analysis

MAUP and the Ecological Fallacy

- Modifiable Areal Unit Problem (MAUP)
- Statistical relations depend on boundaries
- (One) Indicator of the ecological fallacy

MAUP and the Ecological Fallacy



Modifying Spatial Objects

Changing geometries with sf

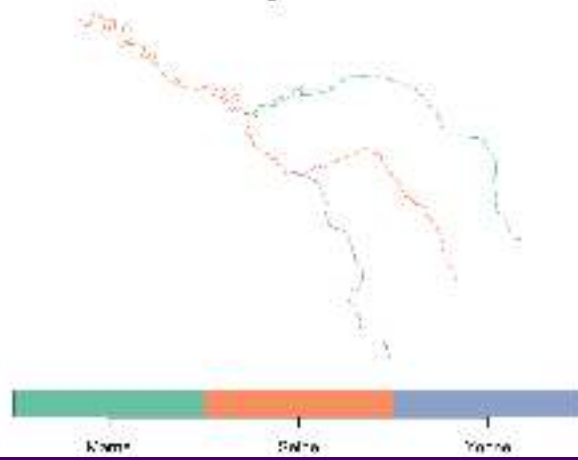
Revisiting predicates and measures

- **Predicates:** evaluate a logical statement asserting that a property is **TRUE**
- **Measures:** return a numeric value with units based on the units of the CRS
- Unary, binary, and n-ary distinguish how many geometries each function accepts and returns

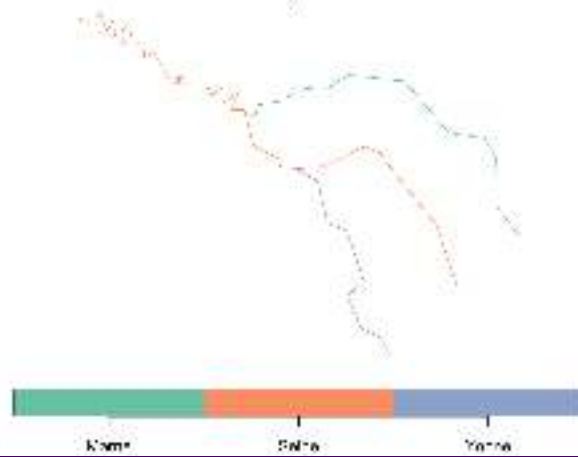
Transformations

- **Transformations:** create new geometries based on input geometries

Original Data



Simplified



Unary Transformations

transformer	returns a geometry ...
<code>centroid</code>	of type POINT with the geometry's centroid
<code>buffer</code>	that is larger (or smaller) than the input geometry, depending on the buffer size
<code>jitter</code>	that was moved in space a certain amount, using a bivariate uniform distribution
<code>wrap_dateline</code>	cut into pieces that do no longer cover the dateline
<code>boundary</code>	with the boundary of the input geometry
<code>convex_hull</code>	that forms the convex hull of the input geometry
<code>line_merge</code>	after merging connecting LINESTRING elements of a MULTILINESTRING into longer LINESTRINGs .
<code>make_valid</code>	that is valid
<code>node</code>	with added nodes to linear geometries at intersections without a node; only works on individual linear geometries
<code>point_on_surface</code>	with a (arbitrary) point on a surface
<code>polygonize</code>	of type polygon, created from lines that form a closed ring

Other Unary Transformations

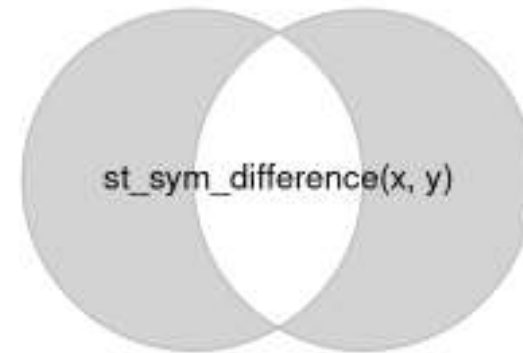
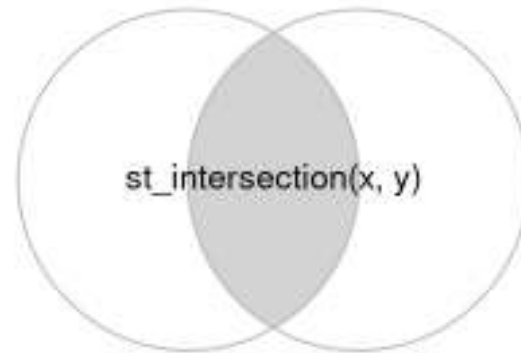
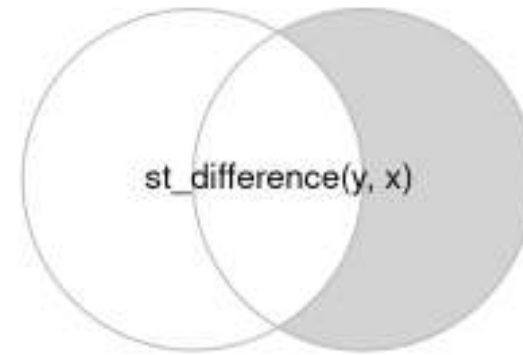
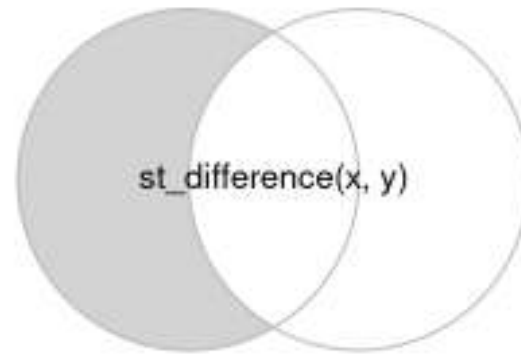
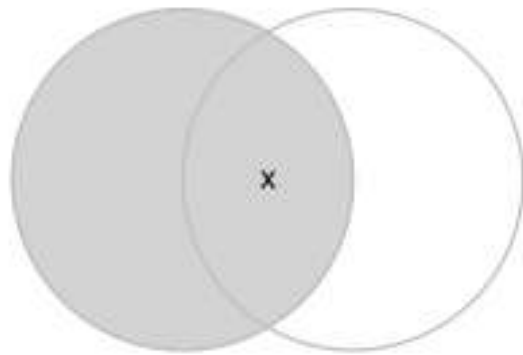
transformer	returns a geometry ...
<code>segmentize</code>	a (linear) geometry with nodes at a given density or minimal distance
<code>simplify</code>	simplified by removing vertices/nodes (lines or polygons)
<code>split</code>	that has been split with a splitting linestring
<code>transform</code>	transformed or convert to a new coordinate reference system (chapter @ref(cs))
<code>triangulate</code>	with Delauney triangulated polygon(s) (figure @ref(fig:vor))
<code>voronoi</code>	with the Voronoi tessellation of an input geometry (figure @ref(fig:vor))
<code>zm</code>	with removed or added Z and/or M coordinates
<code>collection_extract</code>	with subgeometries from a GEOMETRYCOLLECTION of a particular type
<code>cast</code>	that is converted to another type
<code>+</code>	that is shifted over a given vector
<code>*</code>	that is multiplied by a scalar or matrix

Binary Transformers

Binary Transformers

function	returns	infix operator
<code>intersection</code>	the overlapping geometries for pair of geometries	<code>&</code>
<code>union</code>	the combination of the geometries; removes internal boundaries and duplicate points, nodes or line pieces	<code> </code>
<code>difference</code>	the geometries of the first after removing the overlap with the second geometry	<code>/</code>
<code>sym_difference</code>	the combinations of the geometries after removing where they intersect; the negation (opposite) of <code>intersection</code>	<code>%/%</code>
<code>crop</code>	crop an sf object to a specific rectangle	

Binary Transformers



Common Uses of Binary Transformers

- Relating partially overlapping datasets to each other
- Reducing the extent of vector objects

N-ary Transformers

- Similar to Binary (except `st_crop`)
- `union` can be applied to a set of geometries to return its geometrical union
- `intersection` and `difference` take a single argument, but operate (sequentially) on all pairs, triples, quadruples, etc.

Changing geometries with terra

- Changing pixel size with **aggregate** and **disagg** (leaves CRS unchanged)
- Dramatic changes and more sophisticated downsampling with **resample**

